

Distance - time Graphs

Starter:



1. What 2 pieces of information do we need to know to calculate speed?

Distance travelled and time

2. What is the equation for calculating speed?

Speed = distance/ time

3. If an object has moved 300m in 55s, what is the speed of the object?

$$S = d/t$$

$$= 300/ 55 = 5.45 \text{ m/s}$$

4. If a car took 2 hours to travel the distance of 60 km, what was its speed?

$$= 60/ 2 = 30 \text{ km/h}$$

3. Distance-Time Graphs

Learning objectives:

- Plot and interpret a distance-time graph.
- Be able to say what stationary and constant speed looks like on a distance-time graph.

Keywords

- Constant
- Stationary

Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance $\div$ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the $h\rho g$ formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.

Complete this worksheet

Name: _____

Speed: Quick quiz (= a pun!)

1. Convert 50 km/hr to meters per second.

.....

.....

2. What is the speed of a bus that travels 250.0 km in 2.00 hours?

.....

.....

3. A vehicle travels 2345 m in 315 s toward the evening sun. What is its speed and which direction is it traveling(= its velocity)?

.....

.....

4. What distance will a car traveling 65 km/hr travel in 3.0 hrs?

.....



Speed



Time



Distance

Speed: Quick quiz (= a pun!)

1. Convert 50 km/hr to meters per second.

.....
.....

50 km in metres?

= 50,000 metres per hour

Hour to seconds? $60 \times 60 = 3600$

= $50000 / 3600 = 13.9$ m/s

2. What is the speed of a bus that travels 250.0 km in 2.00 hours?

.....
.....

$s = d/t$

= $250/2$

125 kmh

3. A vehicle travels 2345 m in 315 s toward the evening sun. What is its speed and which direction is it traveling(= its velocity)?

.....
.....

$s = d/t$

= $2345 / 315$

= 7.44 m/s

= west (bonus point)

direction?

4. What distance will a car traveling 65 km/hr travel in 3.0 hrs?

.....
.....

$s = d/t$

$d = s \times t$

= 65×3

= 195 km

5. What distance will be traveled if you are going 120km/hr for 30 min?

$$s = d/t$$

$$d = s \times t \quad 30 \text{ mins in hours?} = 0.5 \text{ hr}$$

$$= 120 \times 0.5 \text{ hr}$$

$$= 60 \text{ km}$$

6. How long will it take to go 150 km traveling at 50 km/hr?

$$s = d/t$$

$$t = d/s$$

$$= 150 / 50$$

$$= 3 \text{ hours}$$

7. How long will it take to travel 200 km traveling 100 m/s?

$$s = d/t$$

$$t = d/s$$

$$= 200\,000 / 100 = 2000 \text{ s}$$

200km in m?

2000s in minutes?

8. If a rocket travels 5600 km in 3.00 hours, what is its speed?

$$2000 / 60 = 33.3 \text{ minutes (33 mins 20 s)}$$

$$s = d/t$$

$$= 5600 / 3$$

$$= 1867 \text{ km/h}$$

Motion and the gradient of a distance-time graph

The movement of two objects can be shown on a **distance-time graph** as separate lines.

The speed of each can be found using:

$$\text{speed} = \text{distance} \div \text{time}$$

a

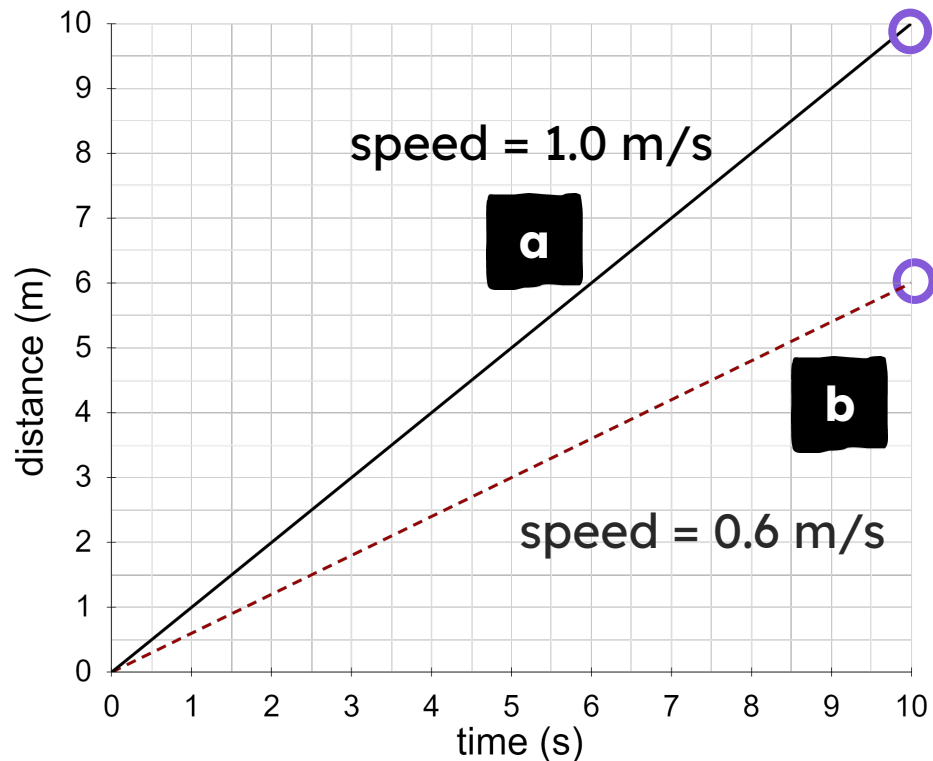
$$\text{speed} = 10 \div 10$$

$$\text{speed} = 1.0 \text{ m/s}$$

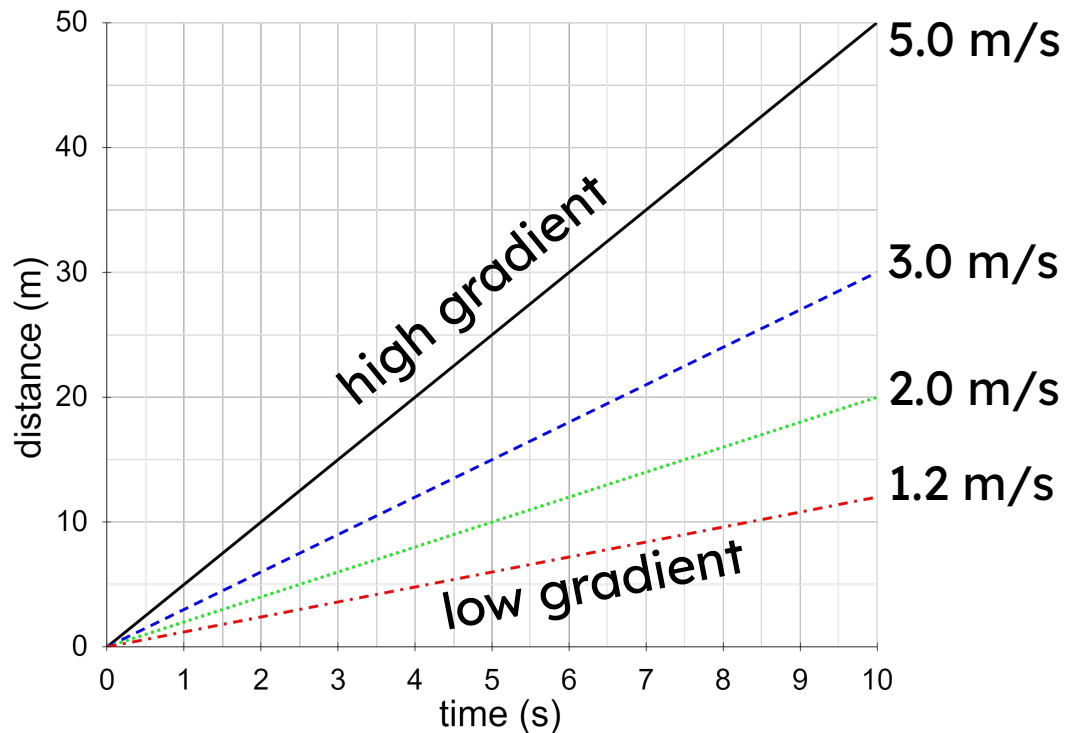
b

$$\text{speed} = 6 \div 10$$

$$\text{speed} = 0.6 \text{ m/s}$$



Motion and the gradient of a distance-time graph



The steeper the line on the graph, the faster the object is moving.

The steepness is called the **gradient**.



Motion and the gradient of a distance-time graph

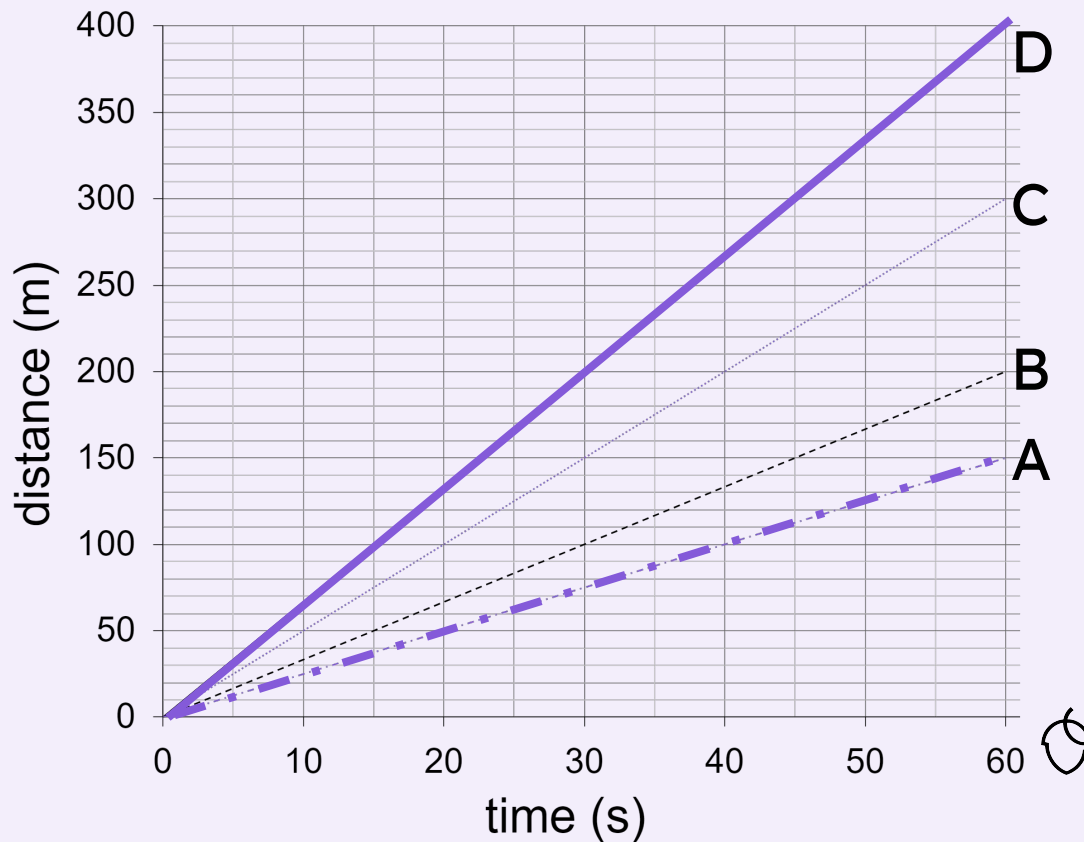


Which object is moving at the highest speed?

D has the highest speed

Which object is moving at the lowest speed?

A has the lowest speed



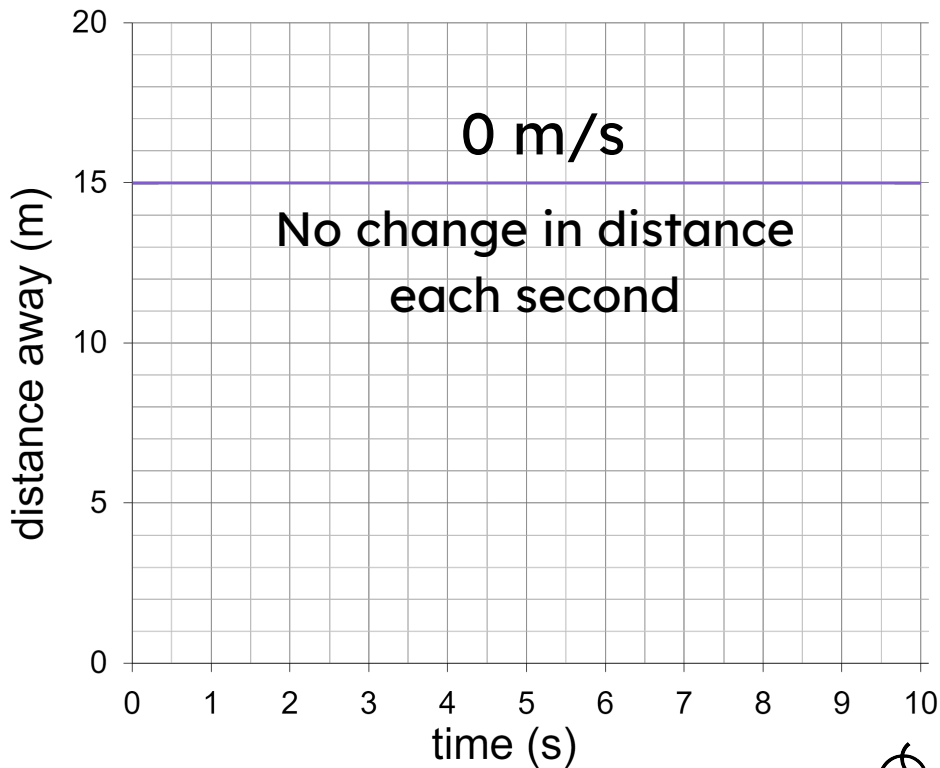
Motion and the gradient of a distance-time graph

A horizontal line on a **distance time graph** shows that the object is not moving.

The distance away changes by 0 m each second.

The speed is 0 m/s.

The object is **stationary**.

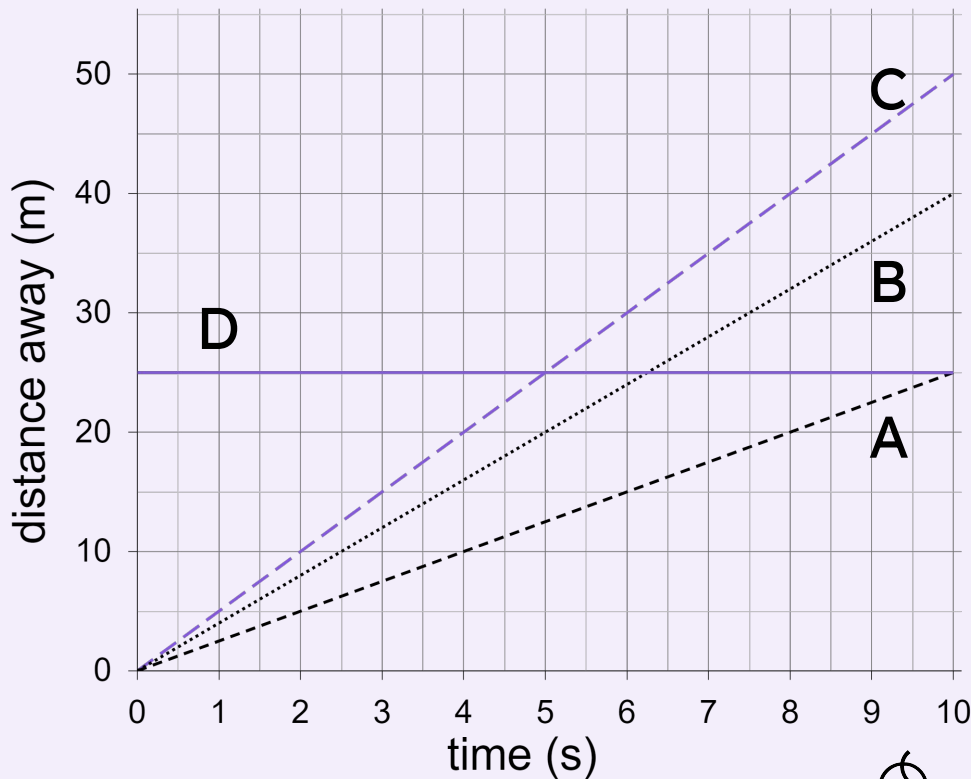


Motion and the gradient of a distance-time graph



Which line shows:

- a) The object that is **stationary**. D
- b) The object that is moving at the highest speed. C
- c) The object that is moving at the lowest speed. A



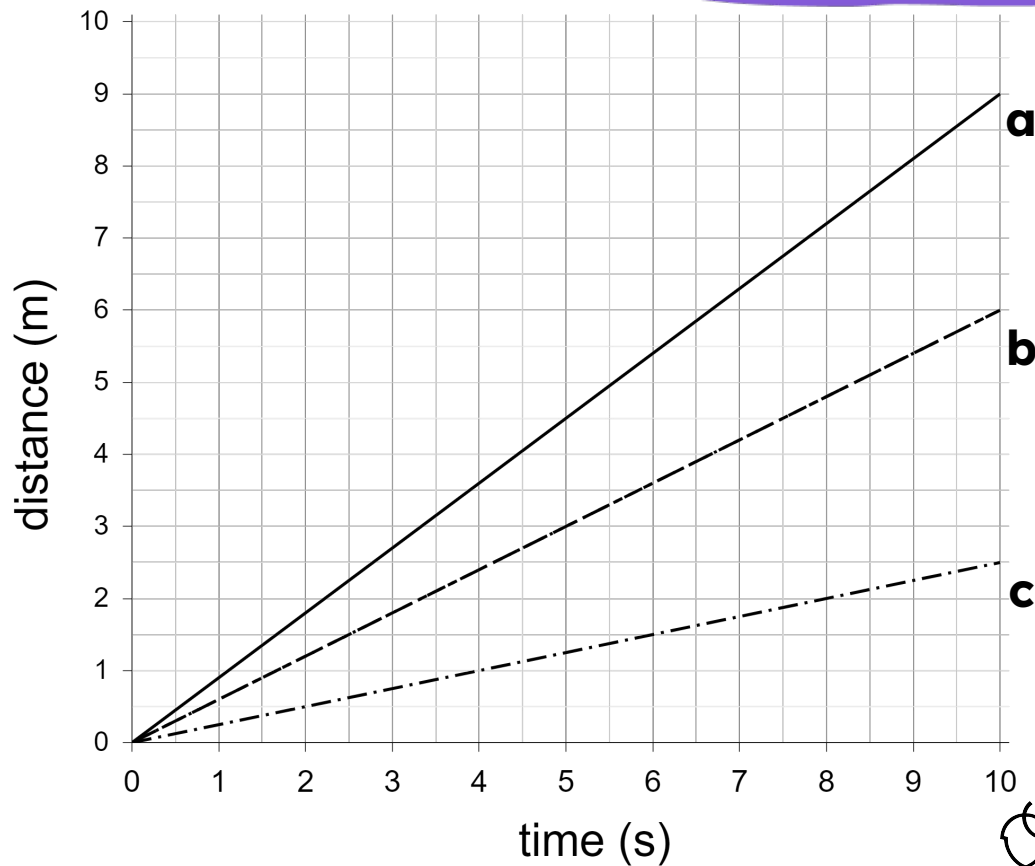
Task A Motion and the gradient of a distance-time graph



Find the **average speeds** of the three objects shown on the graph.



Make sure you include your working on the worksheet itself!



Task A Motion and the gradient of a distance-time graph



Feedback

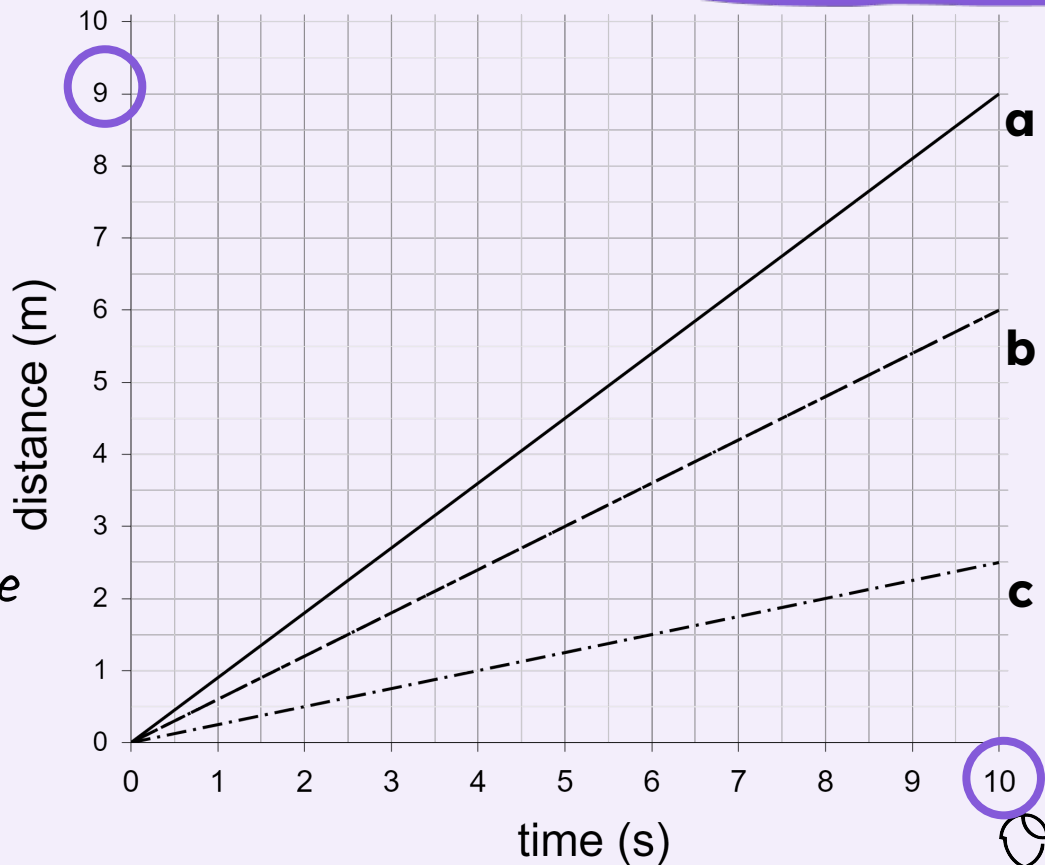
Find the **average speeds** of the three objects shown on the graph.

Line a:

$\text{average speed} = \text{distance} \div \text{time}$

$\text{average speed} = 9 \div 10$

$\text{average speed} = 0.9 \text{ m/s}$



Task A Motion and the gradient of a distance-time graph



Feedback

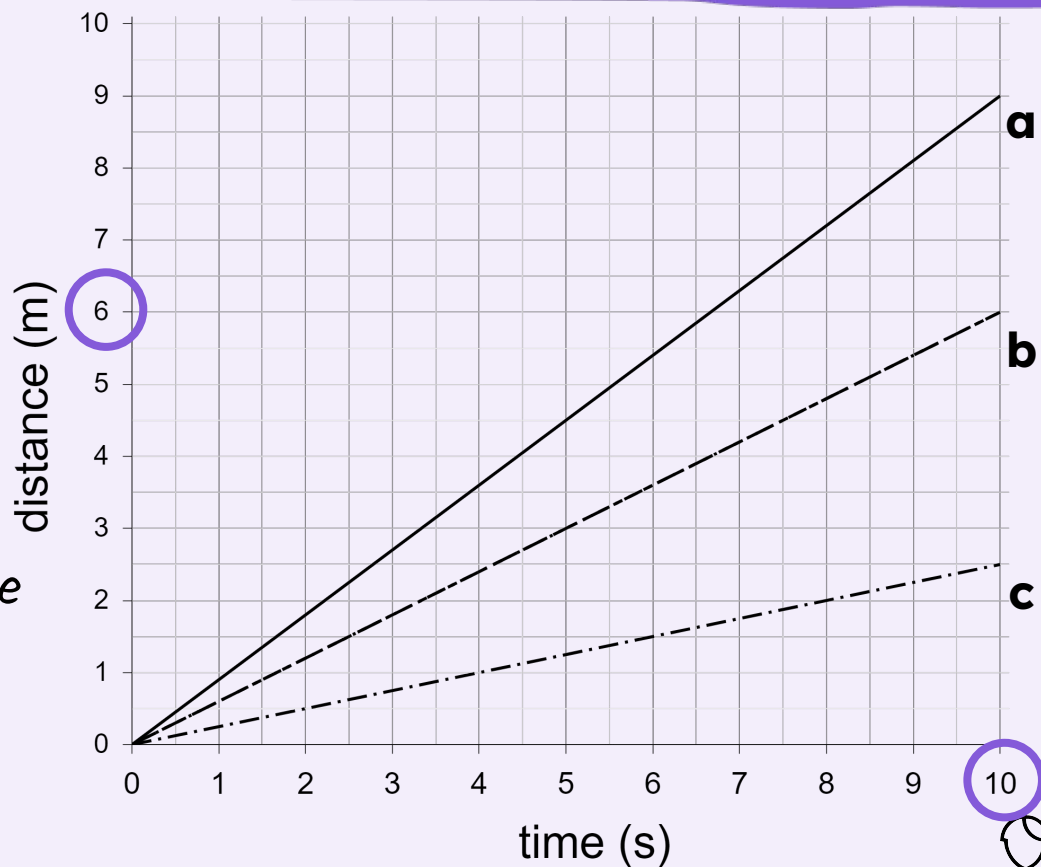
Find the **average speeds** of the three objects shown on the graph.

Line b:

$$\text{average speed} = \text{distance} \div \text{time}$$

$$\text{average speed} = 6 \div 10$$

$$\text{average speed} = 0.6 \text{ m/s}$$



Task A Motion and the gradient of a distance-time graph



Feedback

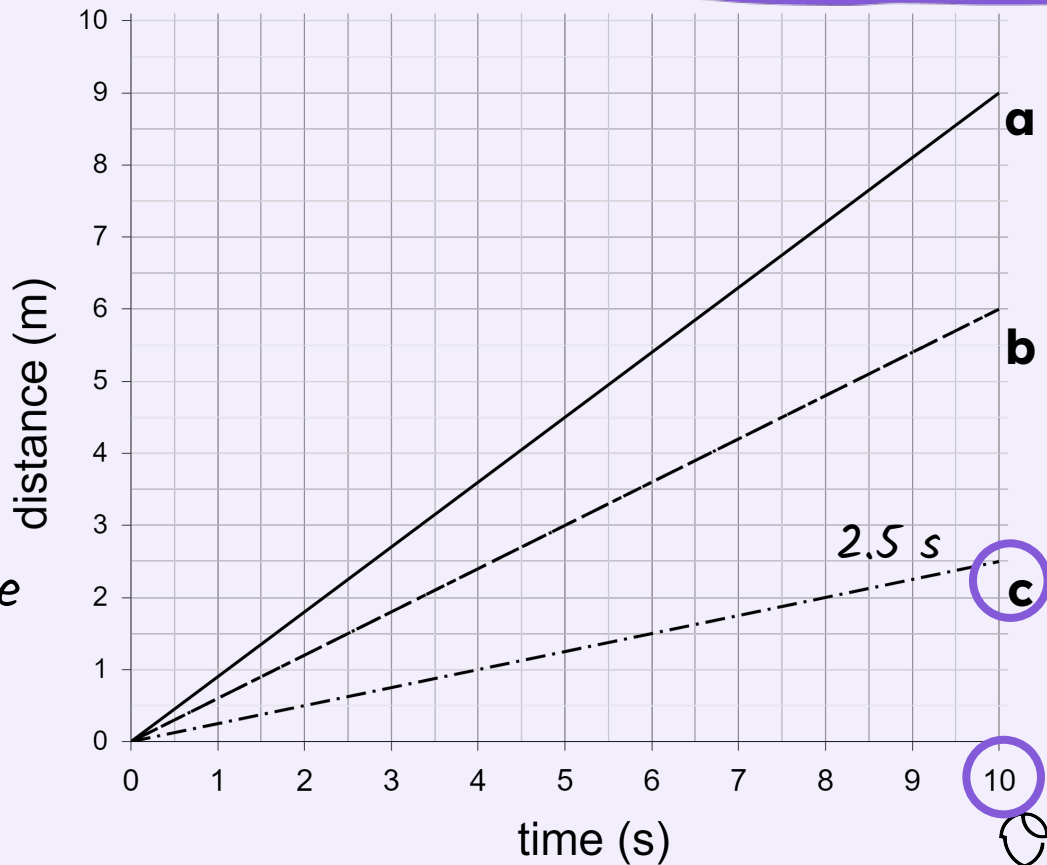
Find the **average speeds** of the three objects shown on the graph.

Line c:

$\text{average speed} = \text{distance} \div \text{time}$

$\text{average speed} = 2.5 \div 10$

$\text{average speed} = 0.25 \text{ m/s}$



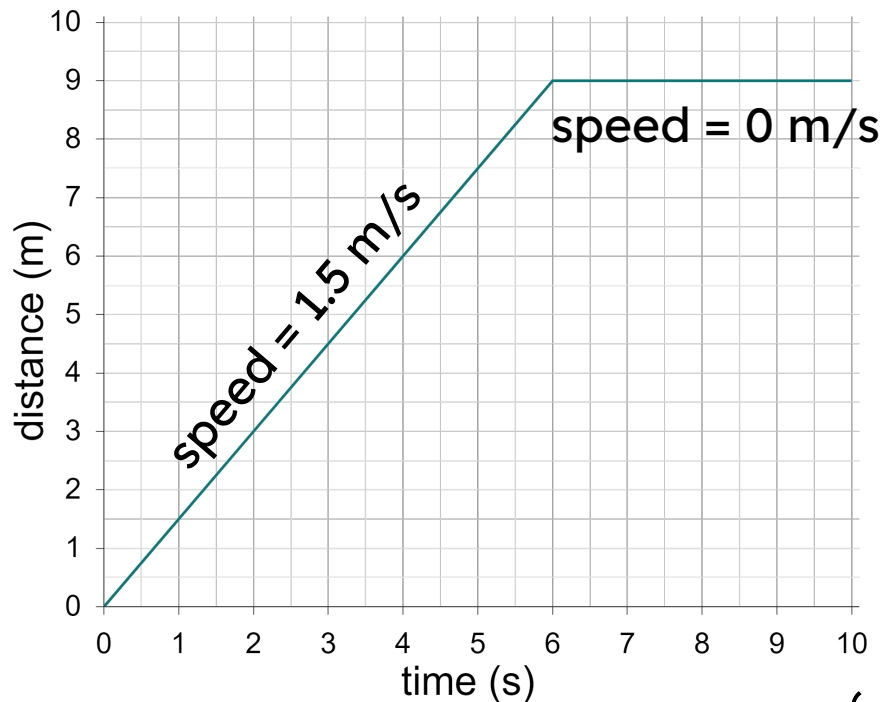
Changing speed on a distance-time graph

The speed of an object can change over time.

This is shown by changes in the **gradient**.

0 - 6 s : speed = 1.5 m/s

6 - 10 s : speed = 0 m/s



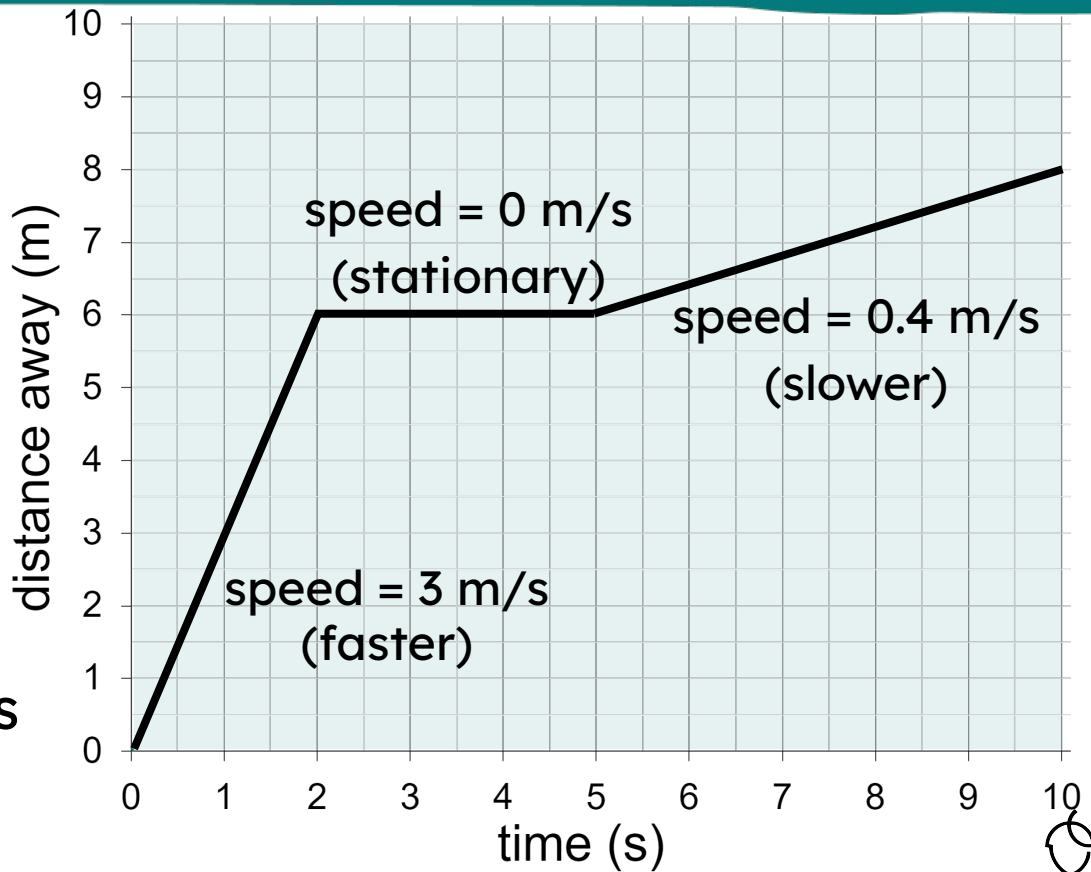
Changing speed on a distance-time graph

Movement can be shown in separate parts or phases.

0 - 2 s : speed = 3 m/s

2 - 5 s : speed = 0 m/s

5 - 10 s : speed = 0.4 m/s



Changing speed on a distance-time graph

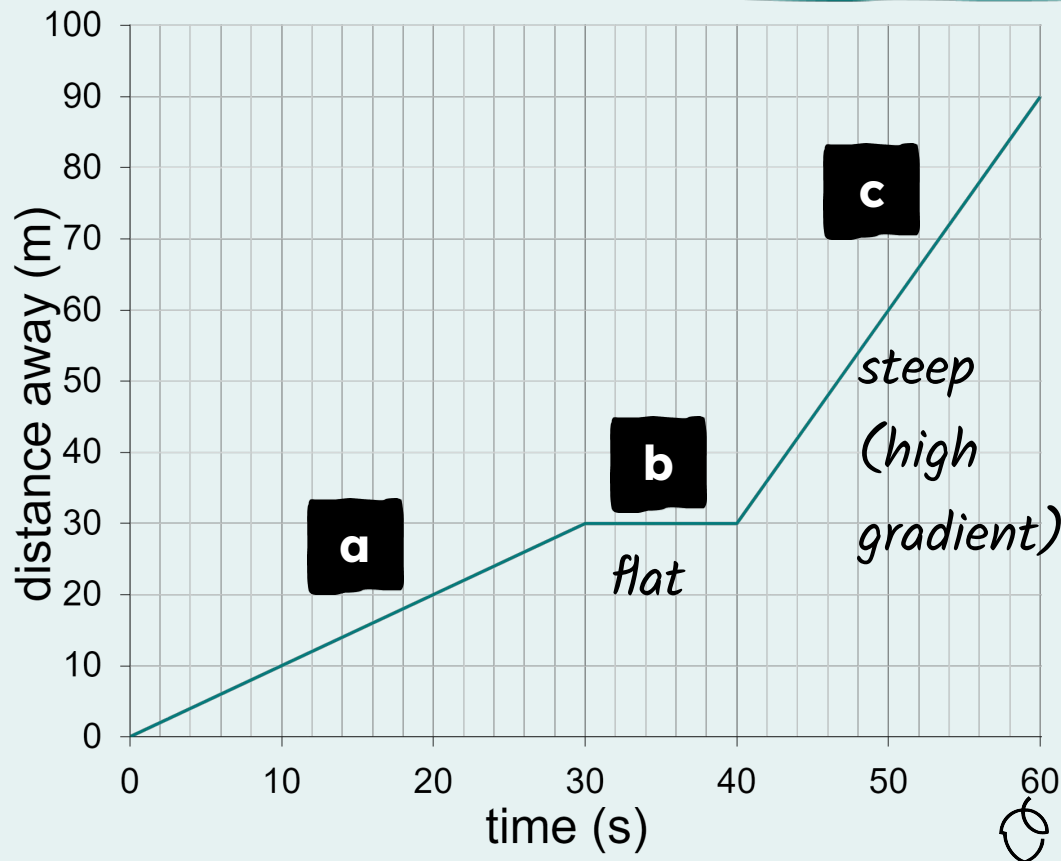


In which part of the movement is the object stationary?

stationary during part B

In which part of the movement is the object moving fastest?

fastest during part C



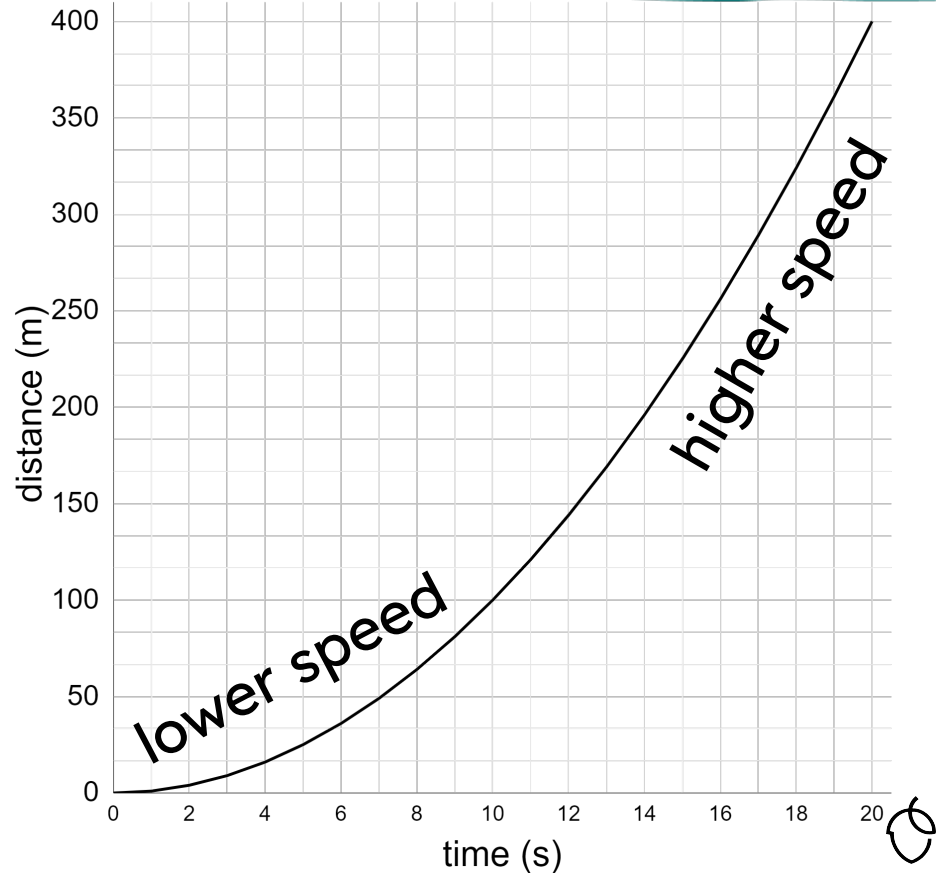
Changing speed on a distance-time graph

A distance-time graph can show that an object is gradually speeding up.

The line is a curve.

The line is steeper with a higher gradient at high speeds.

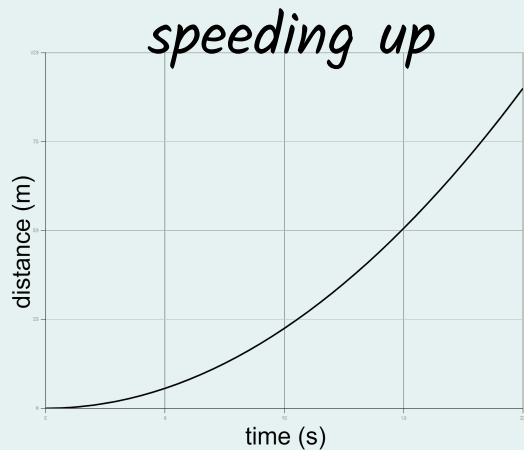
The line is flatter (less steep) with a lower gradient at low speeds.



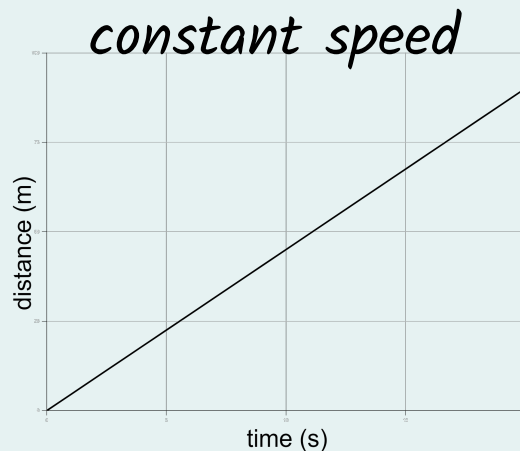
Changing speed on a distance-time graph



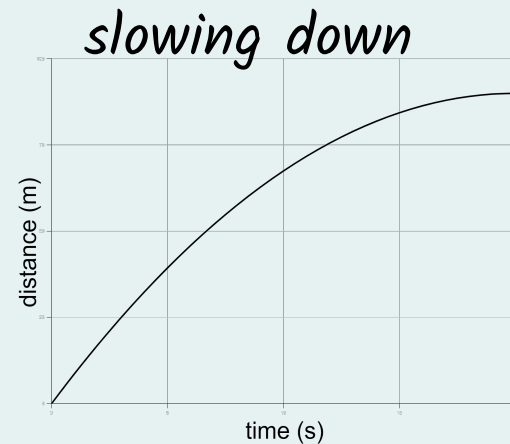
Which of these distance-time graphs shows an object slowing down to a stop?



a



b



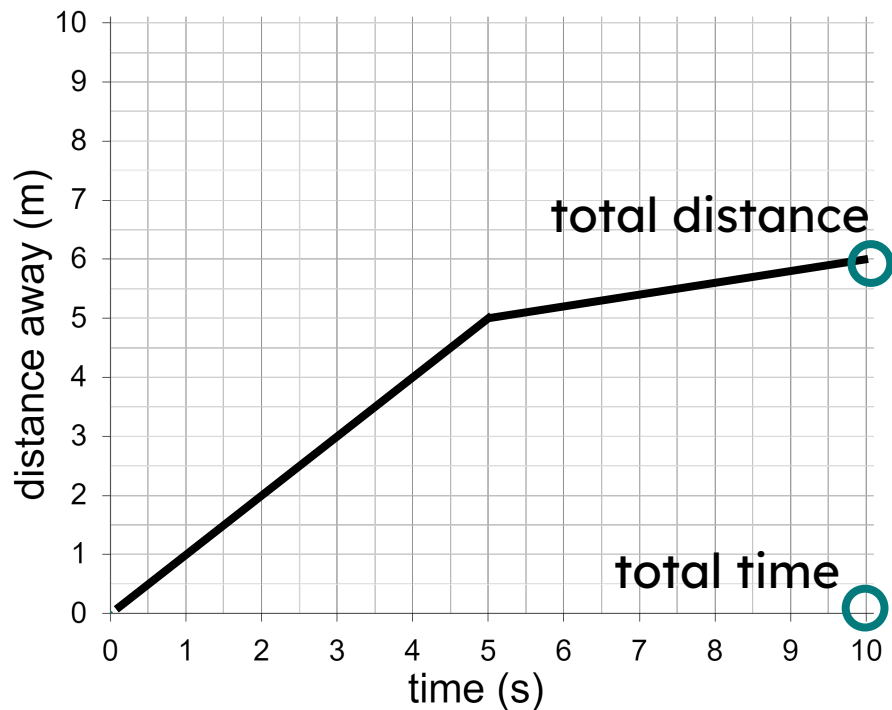
c ✓

Changing speed on a distance-time graph

A graph can tell a story about a journey.



Changing speed on a distance-time graph



The **average speed** of an object for the total journey is found using the total time and the total distance.

$$\text{ave. speed} = \text{total distance} \div \text{total time}$$

$$\text{ave. speed} = 6 \div 10$$

$$\text{ave. speed} = 0.6 \text{ m/s}$$

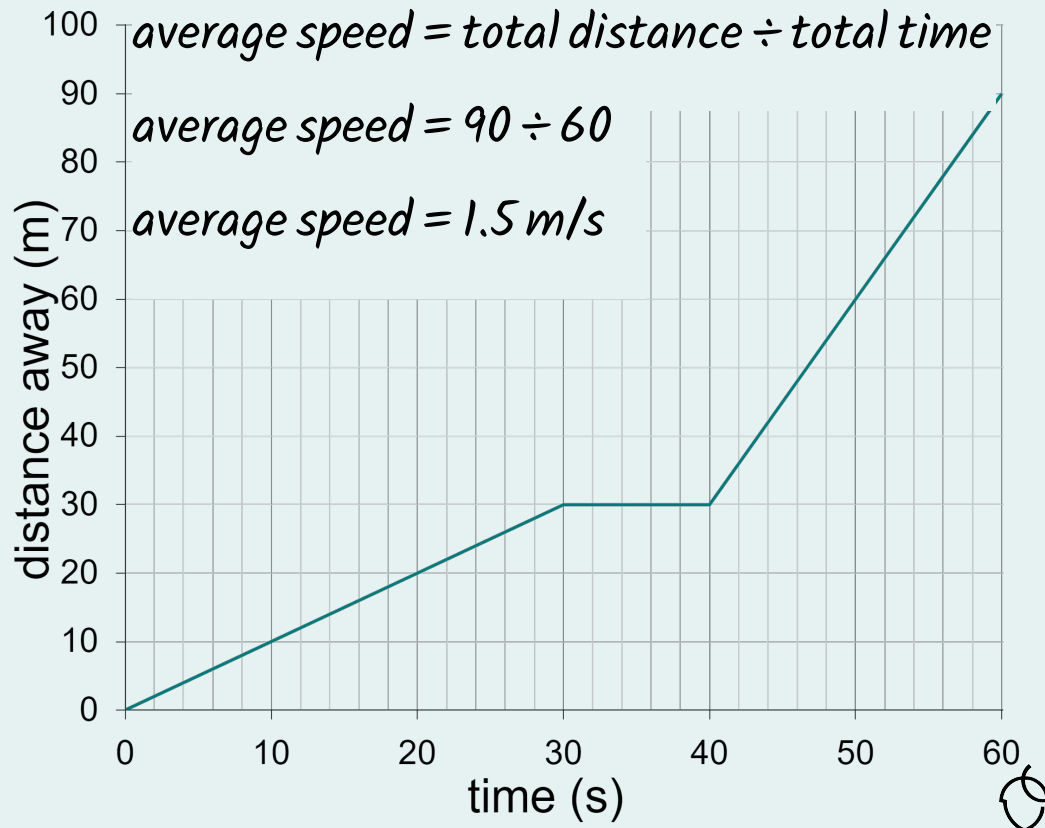


Changing speed on a distance-time graph

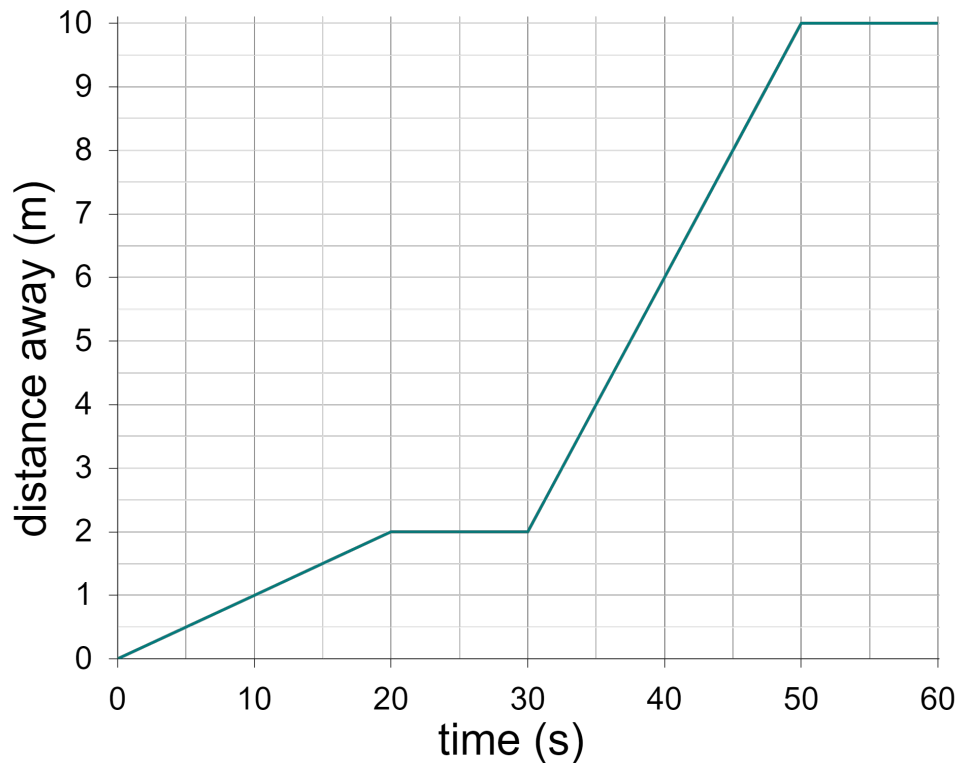


What is the average speed for this complete journey?

- a** 1.0 m/s
- b** 1.5 m/s ✓
- c** 2.0 m/s
- d** 2.5 m/s



Task B Changing speed on a distance-time graph

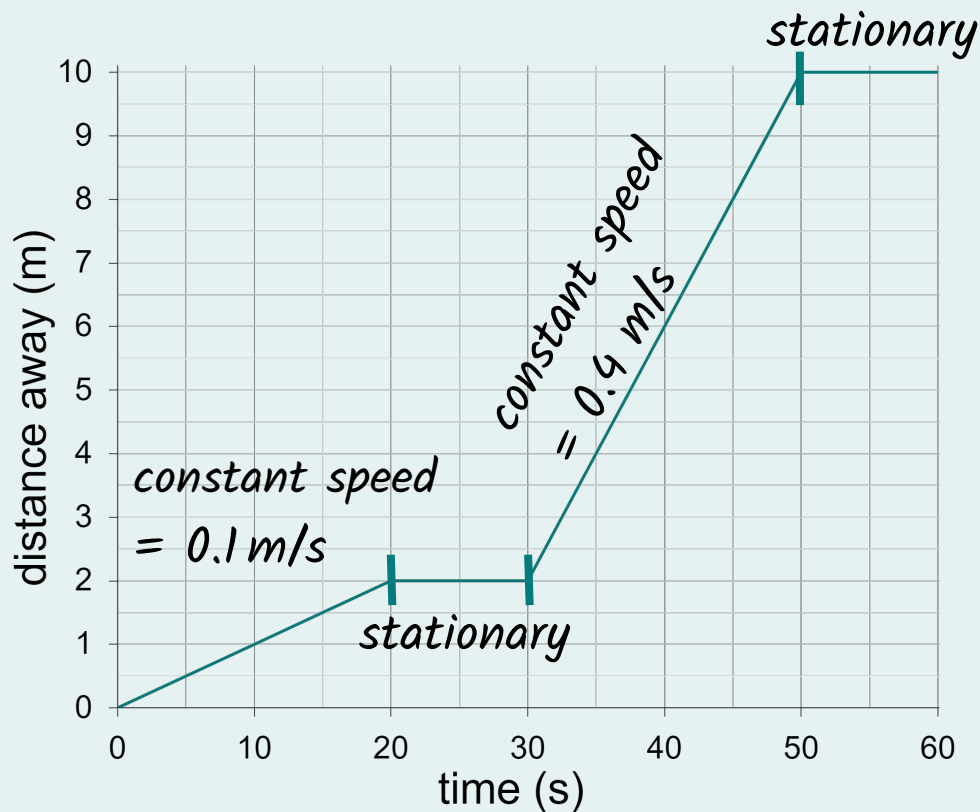


The graph shows the movement of a toy robot.

- 1) Add labels to the graph to describe the movement of the robot in detail.
- 2) Calculate the average speed of the robot during the first 50 s.

Complete this on your worksheet!

Task B Changing speed on a distance-time graph

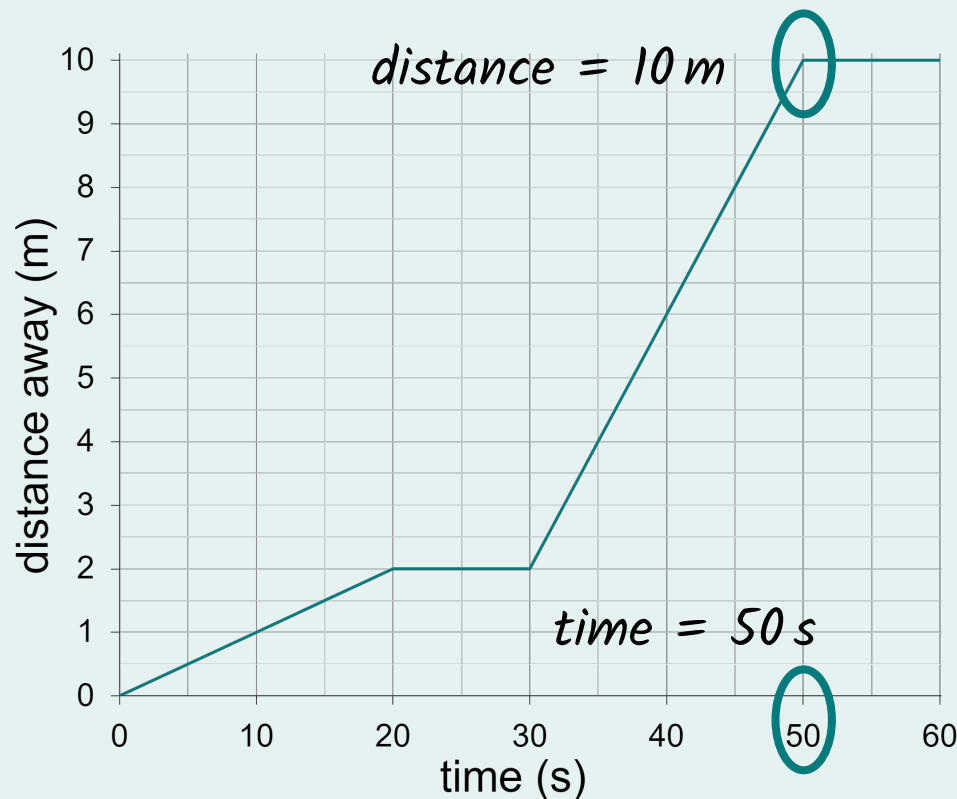


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Task B Changing speed on a distance-time graph



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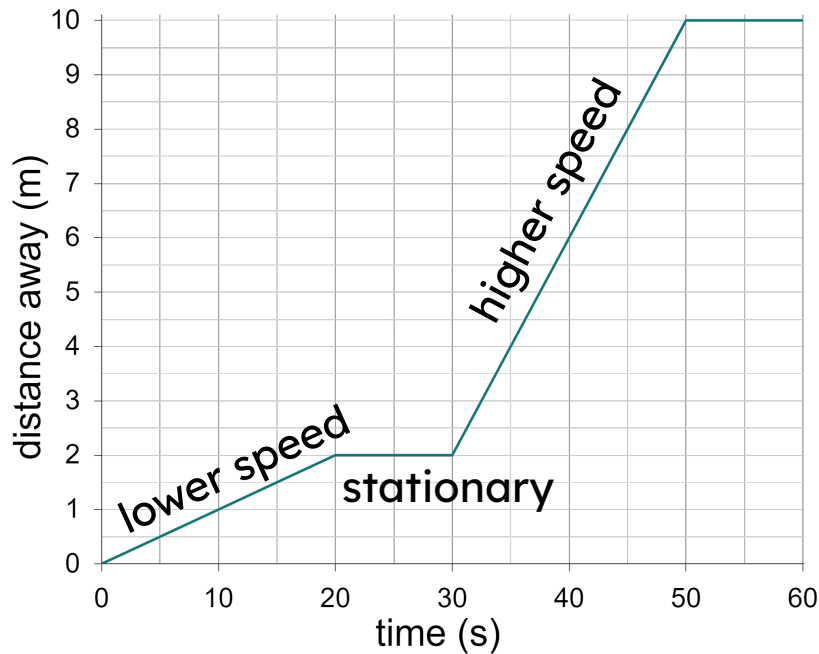
$$\text{average speed} = \text{distance} \div \text{time}$$

$$\text{average speed} = 10 \div 50$$

$$\text{average speed} = 0.2 \text{ m/s}$$



Summary Interpreting distance-time graphs



The steeper the **gradient** the higher the speed.

Horizontal (flat) lines show that the object is **stationary**.

Slopes in opposite directions show forwards and backwards movement.

Speed in any section of the graph is found using:

$$\text{average speed} = \text{change in distance} \div \text{change in time}$$

